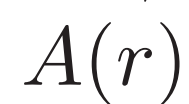
$\mathbf{r} = (r, \phi)$

$$\psi_{\pm\ell} = A(r)\exp(\pm i\ell\phi)$$


$A(r)$

$$\exp(\pm i\ell\phi)$$



$\alpha$

$H_m(\cdot)$

$$\psi_{mn}(z) = (m + n + 1) \tan^{-1}(z/z_0)$$

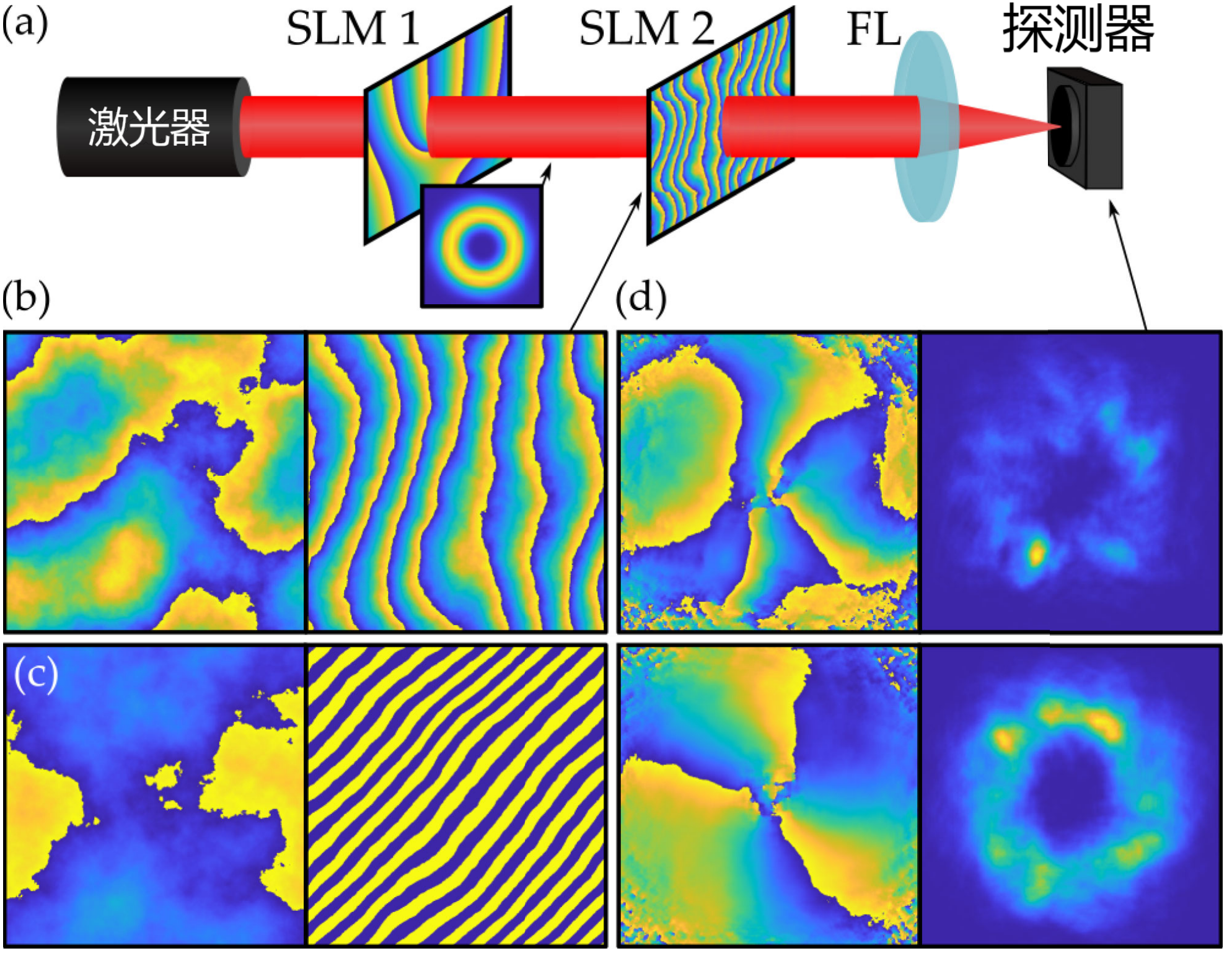
$m = n = 0$

$$\hat{e}_R = \begin{bmatrix} 1 & 0 \end{bmatrix}^\intercal$$

$$\hat{e}_L = \begin{bmatrix} 0 & 1 \end{bmatrix}^\intercal$$

$\ell = 3$

$$\Phi_n^{vK}(\kappa, l_0, L_0) = 0.033 C_n^2 \frac{\exp\left(-\kappa^2/k_m^2\right)}{\left(\kappa^2 + k_0^2\right)^{11/6}} \quad \text{当} \quad 0 \leq \kappa < \infty,$$


(a)
SLM 1
SLM 2
FL
探测器
激光器
(b)
(d)
(c)

| $c$ | $M_1$ | $M_2$ | $N_1$ | $N_2$ | $\sigma^2$ |
|---|---|---|---|---|---|
| 0 | 1.759 | -0.7626 | -1.289 | 0.3166 | 2150 |

[209] M. Krenn, J. Handsteiner, M. Fink, R. Fickler, and A. Zeilinger, "Twisted photon entanglement through turbulent air across vienna," *Proc. Nat. Acad. Sci.*, vol. 112, no. 46, pp. 14 197–14 201, 2015.

[210] J. Davis and W. Tango, "Measurement of the Atmospheric Coherence Time," *Publications Astron. Soc. Pacific*, vol. 108, no. 723, p. 456, May 1996.

[211] A. Rodriguez-Gomez, F. Dios, J. A. Rubio, and A. Comeron, "Temporal statistics of the beam-wander contribution to scintillation in ground-to-satellite optical links: An analytical approach," *Appl. Opt.*, vol. 44, no. 21, Jul. 2005, Art. no. 4574.